Chapter 8. Classification Errors and the Bayes Classifier

Lecture 7-1 (error rate, Bayes optimal classifier, KNN, bias–variance, confusion matrix, ROC/AUC, CV for classification)

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1 The Classification Problem

Accuracy is measured by the **error rate**

$$\text{Error rate} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(y_i \neq \hat{y}_i),$$

Goals: a classifier $C(X)$ with labels in $\mathcal{C}$, the uncertainty of each classification, the roles of the predictors.

2 The Bayes Optimal Classifier

Motivation. Assigning each x to its most probable class gives a lower bound on every method's error.

Classes $1, \ldots, K$; risk $R(C) = \Pr(C(X) \neq Y)$.

Theorem: Bayes classifier

The risk is minimized by

$$C^*(x) = \arg\max_k \Pr(Y = k \mid X = x),$$

assigning each x to the class with the largest conditional probability. Its risk $R(C^*) = 1 - \mathbb{E}\big[\max_k \Pr(Y = k \mid X)\big]$ is the **Bayes error rate**, the lowest error achievable by any classifier.

Proof [Supplement]. $R(C) = 1 - \Pr(C(X) = Y) = 1 - \mathbb{E}\big[\Pr(Y = C(X) \mid X)\big]$. For each x, $\Pr(Y = C(x) \mid X = x) \leq \max_k \Pr(Y = k \mid X = x)$, with equality when $C(x) = C^*(x)$. Taking expectations, $R(C) \geq R(C^*)$. ■

The Bayes error is positive whenever the classes overlap. **Bayes decision boundary**: $\{x : \Pr(Y = 1 \mid x) = 0.5\}$.

3 Approximating the Bayes Classifier

Test error of $\hat{C}$:

$$\text{Err}_{\text{Te}} = \text{Ave}_{i \in \text{Te}}\, \mathbb{I}\{y_i \neq \hat{C}(x_i)\}.$$

Either *estimate the conditional probabilities* and plug in (logistic regression, GAMs, LDA/QDA, KNN), or *model* $C(x)$ *directly* (support-vector machines).

3.1 K-nearest neighbors

$$\hat{\Pr}(Y = k \mid X = x) = \frac{1}{K}\sum_{x_i \in \mathcal{N}_K(x)} \mathbb{I}(y_i = k),$$

and predict the majority class. **Smaller K, more complex**: $K = 1$ is overly flexible (training error 0), $K = 100$ nearly linear, $K = 10$ close to the Bayes boundary.

3.2 Bias–variance in classification

As $1/K$ grows, training error declines; test error is U-shaped.

4 Other Measures for Classification

Motivation. With rare classes the overall error rate can look excellent for a useless classifier.

4.1 Confusion matrix

Default data (LDA fit, $n = 10{,}000$):

		True default status No	Yes	Total
Predicted	No	9,644	252	9,896
	Yes	23	81	104
	Total	9,667	333	10,000

- Overall error: $(23 + 252)/10000 = 2.75\%$ — but always predicting "No" would already give $333/10000 = 3.33\%$.
- **False positive rate**: fraction of negatives classified positive, $23/9667 = 0.2\%$.
- **False negative rate**: fraction of positives classified negative, $252/333 = 75.7\%$.

Sensitivity (recall, TPR) $= 1 - \text{FNR} = 81/333 = 24.3\%$; specificity $= 1 - \text{FPR} = 99.8\%$: unacceptable for catching defaulters.

4.2 Thresholds

Threshold 0.5 mimics the Bayes rule; lowering it (0.2, 0.1) trades fewer false negatives for more false positives. With costs c_{FN}, c_{FP}: predict "Yes" iff $\hat{p}\, c_{FN} > (1 - \hat{p}) c_{FP}$, i.e. $\hat{p} > c_{FP}/(c_{FP} + c_{FN})$ (Exercise 8.1).

4.3 ROC curve and AUC

ROC curve: TPR against FPR as the threshold moves from 1 to 0. **AUC**: area under it; 0.5 is random guessing, 1 perfect separation.

Theorem: probabilistic meaning of AUC

For a score $s(x)$, $\text{AUC} = \Pr\left(s(X^+) > s(X^-)\right)$ for independent draws X^+ from the positive and X^- from the negative class (ties counted 1/2).

Proof [Supplement]. With threshold t, $\text{TPR}(t) = \Pr(s^+ > t)$ and $\text{FPR}(t) = \Pr(s^- > t)$. Then $\text{AUC} = \int \text{TPR}\, d\text{FPR} = \int \Pr(s^+ > t)\, dF_-(t) = \Pr(s^+ > s^-)$. ■

AUC depends only on the ranking: invariant to monotone transformations and class proportions.

5 Cross-Validation for Classification

With parts $C_1, \ldots, C_K$ of sizes n_k,

$$\text{CV}_K = \sum_{k=1}^{K} \frac{n_k}{n} \text{Err}_k, \qquad \text{Err}_k = \frac{1}{n_k} \sum_{i \in C_k} \mathbb{I}(y_i \neq \hat{y}_i),$$

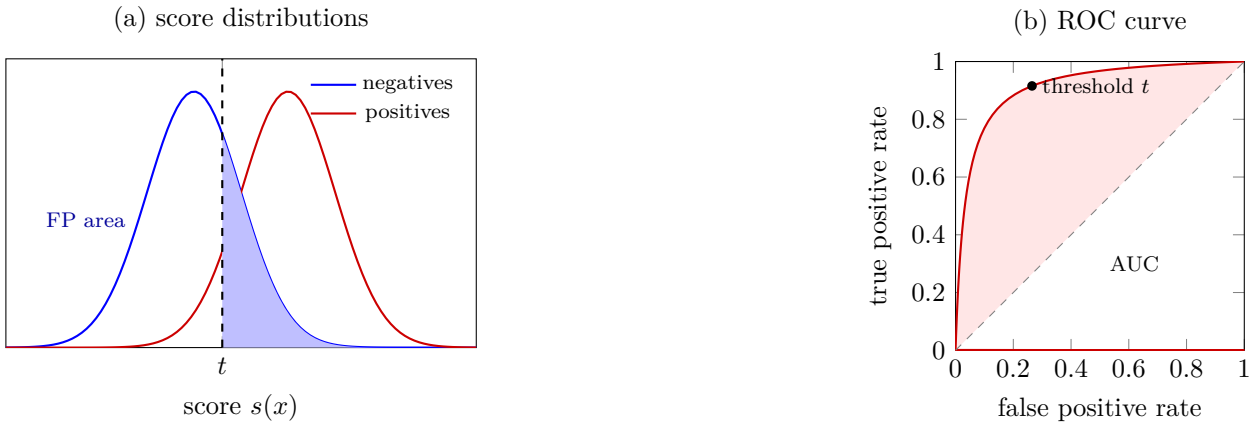


Figure 1: (a) Score distributions; shaded: FPR at threshold t. (b) Moving t traces the ROC curve.

$\hat{y}_i$ fitted without part k. For polynomial logistic regression and KNN on simulated data, 10-fold CV may underestimate the test error level but its minimum is near the right complexity.

Summary

- The Bayes classifier $\arg\max_k \Pr(Y = k \mid x)$ attains the lowest error; KNN votes locally.
- Report FPR, FNR, sensitivity, specificity; choose thresholds by costs; AUC $= \Pr(s^+ > s^-)$.

6 Selected exercises

Motivation. The cost-weighted threshold and the two-Gaussian Bayes error used in Ch. 10.

Exercise 8.1 (The cost-sensitive Bayes rule)

Let $Y \in \{0, 1\}$, $\eta(x) = \Pr(Y = 1 \mid X = x)$, and let a classifier C incur cost $c_{FN} > 0$ when $C(X) = 0$, $Y = 1$ and $c_{FP} > 0$ when $C(X) = 1$, $Y = 0$. Show that the expected cost

$$R(C) = c_{FN} \Pr(C(X) = 0, Y = 1) + c_{FP} \Pr(C(X) = 1, Y = 0)$$

is minimized by $C^*(x) = \mathbb{I}\{\eta(x) > c_{FP}/(c_{FP} + c_{FN})\}$, which is the Bayes classifier when $c_{FP} = c_{FN}$.

Solution [Supplement].

Step 1. Conditioning on X,

$$R(C) = \mathbb{E}\big[c_{FN}\, \eta(X) \mathbb{I}\{C(X) = 0\} + c_{FP}\{1 - \eta(X)\}\mathbb{I}\{C(X) = 1\}\big].$$

Step 2. For each x the integrand is $c_{FN}\eta(x)$ if $C(x) = 0$ and $c_{FP}\{1 - \eta(x)\}$ if $C(x) = 1$, so it is at least $\min\{c_{FN}\eta(x), c_{FP}(1 - \eta(x))\}$, the value attained by C^*: $C^*(x) = 1$ exactly when $c_{FP}(1 - \eta) < c_{FN}\eta$, i.e. $\eta > c_{FP}/(c_{FP} + c_{FN})$ (at equality both choices cost the same).

Step 3. Taking expectations, $R(C) \geq R(C^*)$. Equal costs: threshold $1/2$; $c_{FN} = 9c_{FP}$: threshold 0.1 (Default data, Ch. 10). ■

Exercise 8.2 (Bayes error of two Gaussian classes)

Let $\pi_1 = \pi_2 = \frac{1}{2}$ and $X \mid Y = k \sim N(\mu_k, \Sigma)$ in $\mathbb{R}^p$, $\Sigma \succ 0$, $\mu_1 \neq \mu_2$, and $\Delta^2 = (\mu_2 - \mu_1)^\top \Sigma^{-1}(\mu_2 - \mu_1)$ (Mahalanobis distance).
(a) Show that the Bayes classifier assigns class 2 iff $w^\top x > w^\top(\mu_1 + \mu_2)/2$, with $w = \Sigma^{-1}(\mu_2 - \mu_1)$.
(b) Show that the Bayes error is $\Phi(-\Delta/2)$. Check the value $\Phi(-1.5) \approx 6.7\%$ for $p = 1$, $\mu_{1,2} = \mp 1.5$, $\sigma = 1$.

Solution [Supplement].

(a) With equal priors the Bayes rule picks the larger density. The normalizing constants are equal, so

$$\log \frac{f_2(x)}{f_1(x)} = \frac{1}{2}\{(x-\mu_1)^\top \Sigma^{-1}(x-\mu_1) - (x-\mu_2)^\top \Sigma^{-1}(x-\mu_2)\} = w^\top x - \frac{1}{2} w^\top(\mu_1 + \mu_2),$$

where the terms $x^\top \Sigma^{-1} x$ cancel and $\mu_2^\top \Sigma^{-1}\mu_2 - \mu_1^\top \Sigma^{-1}\mu_1 = (\mu_1 + \mu_2)^\top \Sigma^{-1}(\mu_2 - \mu_1)$ by symmetry of Σ^{-1}. Class 2 wins iff this is positive.

(b)

Step 1. Put $c = w^\top(\mu_1 + \mu_2)/2$. Under class 1, $w^\top X \sim N(w^\top \mu_1, w^\top \Sigma w)$ with $w^\top \Sigma w = \Delta^2$, and $c - w^\top \mu_1 = \frac{1}{2} w^\top(\mu_2 - \mu_1) = \Delta^2/2$. Hence $\Pr(w^\top X > c \mid Y = 1) = 1 - \Phi(\Delta/2) = \Phi(-\Delta/2)$.

Step 2. Under class 2, $w^\top \mu_2 - c = \Delta^2/2$, so $\Pr(w^\top X \leq c \mid Y = 2) = \Phi(-\Delta/2)$.

Step 3. The Bayes error is $\frac{1}{2}\Phi(-\Delta/2) + \frac{1}{2}\Phi(-\Delta/2) = \Phi(-\Delta/2)$. For $p = 1$, $\Delta = |\mu_2 - \mu_1|/\sigma = 3$, giving $\Phi(-1.5) = 0.0668$. ■